

1). Water Jug problem

1. Fill the 3-gallon jug
 2. Pour it into the 4-gallon jug ($4G=3$, $3G=0$)
 3. Fill the 3-gallon jug again
 4. Pour water into the 4-gallon jug until it is full
(Only 1 gallon is transferred) Now: $4G=4$, $3G=2$
 5. Empty the 4-gallon jug
 6. Pour the remaining 2 gallons from the 3-gallon jug into the 4-gallon jug
- Result: The 4-gallon jug contains exactly 2 gallon.

2). Mars Rover Agent

i). percepts

Camera images
Rock and soil samples
Temperature
Atmospheric pressure

iii). Action

Move forward/backward
Turn left/right
Collect rock/soil sample
Analyse samples
Capture image

vi). Suitable Agent Architecture

It learns from experience, makes intelligence decision.

ii). Operating Environment:

partially observable
Stochastic (uncertain)
Sequential
Dynamic

iv). performance Measure:

Accuracy of sample analysis
Number of successful mission
Safe navigation
Energy efficiency

3).

8 - Queen Problem

Initial State : Empty 8x8 chessboard

State Space : All possible placement of queens

Operators : place one queen in a safe position
in the next column.

Goal State : Eight queens placed with no two
attacking each other.

Path Cost : Equal cost for each move.

Search Strategy :

- > Backtracking Search (Depth-First Search)
- > Constraint Satisfaction problem (Csp)

4).

OLA Cab Booking System:

Agent Type:

* Goal-Based Agent

Reason:

The agent selects the best cab type based on
the customer's destination, comfort and availability.

Pseudo Code:

```
Input pickup location
Input destination
Display available cab types
User selects cab
check availability
If available
    Calculate fare
    Confirm booking
Else
    Display "Cab not available"

END.
```

5). Uniform Cost Search:

Uniform Cost Search (UCS) is an uninformed search algorithm that expands the node with the lowest cumulative path cost from the start node. It guarantees the optimal (least-cost) solution when all edge costs are positive.

Steps:

1. Start from the source node S .
2. Add neighbouring nodes to the priority queue with their path costs.
3. Select the node with the minimum cost.
4. Repeat until the goal node G is reached.
5. The first path found to the goal is the least-cost (optimal path).

Result:

Dijkstra's algorithm guarantees the shortest path when all the edge costs are positive.